MIL-C-21883D(NU)
1 October 1985
SUPERSEDING
MIL-C-21883C(SA)
16 MARCH 1977

MILITARY SPECIFICATION

CLOTH, SATIN OR CLOTH TWILL, RAYON WARP AND COTTON FILLING

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a satin or twill cloth with a rayon warp and a cotton filling.

1.2. Classification The cloth shall be of one type only and of the following colors (see 6.2):

White	- 3052
Blue	- 3360
Green	- 3451
Yellow	- 3551
Brown	- 3752
Red	- 3851
Purple	- 3907

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1. Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should by obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Goverment documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATION UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580).

* 2.2 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard sample The finished cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Material

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* 3.3.1 Yarn The warp yarn shall be 100 (minimum) denier bright rayon. The filling yarn shall be combed cotton evenly spun into singles or plied yarns (see 4.4.1).

* 3.4 Physical requirements The finished cloth shall conform to the requirements in table I when tested as specified in 4.5.

Table I - Physical requirements

Weight, oz/sq yd (g/m^2) (min)	Yarns per inch (2.54 cm) min		Breaking Strength lbs (newton) min	
	Warp	Filling	Warp	Filling
7.0 (237)	250	70	160 (712)	60 (267)

3.4.1 Weave The weave shall be three up one down (3/1) right hand twill or an eight or five harness warp satin with a count of five and three respectively when tested as specified in 4.5.

* 3.4.2 Width The minimum acceptable width of the finished cloth shall be 40 inches (101.60 cm) exclusive of selvages for both fly shuttle and shuttleless looms, unless otherwise specified (see 6.2).

3.5 Color The color shall be as specified (see 1.2 and 6.2) and shall match the standard sample. When white is specified, the cloth shall be chemically bleached and shall be supplemented with optical brighteners which fluoresce to the blue region.

* 3.5.1 Colorfastness The dyed and finished cloth shall show colorfastness (discoloration) to perspiration, laundering, light and crocking equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available the dyed and finished cloth must meet the minimum adjective requirements of Table II as specified in 4.5.

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Table II - Colorfastness requirements

Colorfastness to: (Discoloration)	Light	Laundering	Perspiration 1/	Crocking Dry	Crocking Wet
Blue-3360	good	good	fair	good	good
Green-3451	fair	fair	fair	good	good
Yellow-3551	good	good	fair	good	good
Brown-3752	fair	good	fair	good	good
Red-3851	good	good	fair	good	good
Purple-3907	good	good	fair	good	good
White-3052	fair	good	good	good	good

1/ Report the results of whichever specimen, acid or alkaline, that has the lowest adjective rating.

* 3.5.2 Matching The color of the finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7000 ± 500 Kelvin and shall be a good approximation to standard sample under incandescent lamplight at 2850 ± 100 Kelvin.

3.5.3 Sulfur content The use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When no standard sample is available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.6 Finish The cloth shall be closely singed, desized and mercerized and shall have a clear natural finish with minimum warpwise tension (see 4.4.1).

3.6.1 Nonfibrous materials The starch and protein content, including chloroform-soluble and water-soluble material, shall not exceed 5.0 percent when tested as specified in 4.5.

3.6.2 Dimensional stability The finished cloth shall not shrink or elongate more than 8.0 percent in the direction of the warp nor more than 3.0 percent in the direction of the filling when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

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3.7 pH The pH value of the water extract of the finished cloth shall be not less than 5.5 nor more than 8.5 when tested as specified in 4.5.

3.8 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be delivered on rolls, in continuous lengths each not less than 40 yards (36.4 m). Each length shall be put-up in full width rolls as specified in PPP-P-1134.

3.9 Fiber identification Each roll of cloth shall be labeled or ticketed, for fiber content in accordance with the Rules and Regulations under the Textile Fiber Product Identification Act.

3.10 Face marking The face side of the cloth shall be identified by applying a stamping on the side of the cloth with the word "FACE" at each end of the piece.

3.11 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (83.6 m^2), when calculated as specified in Section 4, shall not exceed the established point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2. Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3.)
2. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article of finished cloth shall be visually examined for color, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

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4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be visually examined and tested in accordance with all requirements of reference specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the Contractor shall furnish a certificate of compliance for the yarn requirements stated in 3.3.1, and the finishing requirements in 3.6.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4 m) only 100 yards (91.4 m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance 3 feet (0.91 m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "D" for slubs and "C" for knots, shall be scored. No linear yard (increments of 1 yard (0.91 m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 point. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards (83.6 m^2) of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards (83.6 m^2) of two or more individual rolls exceed 45.0 points. If one roll exceeds 45.0 points per 100 square yards (83.6 m^2), a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45.0 points per 100 square yards (83.6 m^2). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 Square yards (83.6 } m^2\text{)}$$

4.4.2.1.1 Demerit points Demerit points shall be assigned as follows:

For defects 3 inches (7.6 cm) or less in any dimension - one point
For defects exceeding 3 inches (7.6 cm), but not exceeding 6 inches (15.2 cm) in any dimension - two points
For defects exceeding 6 inches (15.2 cm), but not exceeding 9 inches (22.9 cm) in any dimension - three points
For defects exceeding 9 inches (22.9 cm) in any dimension - four points

The following defects, when present, shall be scored 4 points for each yard (0.91 m) in which they occur:

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Baggy, ridgy or wavy cloth
Width less than specified
Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
Excessive neppiness

4.4.2.2 Examination for length

4.4.2.2.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards (1.82 m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, identification of preshrinkage process and non-compliance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinkage process by name or trade mark on the cloth or ticket, or two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with section 5 requirements of this specification.

* 4.5 Testing of the end item The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified apply to the results of the determination made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for the test purposes shall be 3 1/2 continuous yards (3.19 m) full width of the finished cloth. The lot size shall be expressed in units of one yard (0.91 m). The lot shall be unacceptable if one or more sample units fail to meet any of the tests requirements specified.The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table III- Test Method

Characteristic	Requirement paragraph	Test Method
Weight	3.4	5041
Yarns per inch		
Warp and Filling	3.4	5050
Breaking strength		
warp and Filling	3.4	5100
Weave	3.4.1	Visual 1/
Colorfastness (Discoloration) to:		
Laundering	3.5.1	5610 2/
Light	3.5.1	5660
Crocking	3.5.1	5651
Perspiration	3.5.1	5680 4/
Sulfur content	3.5.3	2020
Non fibrous material	3.6.1	2611
Shrinkage	3.6.2	5556 3/
pH	3.7	2811

1/One determination per sample unit and the results reported as "Pass" or "Fail".

2/Only the stain on the cotton and rayon fibers of the transfer cloth shall be evaluated.

3/The cotton laundering procedure shall be used.

4/ Report the results of whichever specimen, acid or alkalene, that has the lowest adjective rating.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial The cloth shall be put up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, or Commercial The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

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6.NOTES

6.1 Intended use The cloth covered by this specification is intended for use in flight deck clothing for personnel of the Department of the Navy.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Color of cloth required (see 1.2).
c. Width of cloth if other than specified (see 3.4.2).
d. When first article is required (see 3.2).
e. Minimum length, if other than specified (see 3.7).
F. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Standard sample For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817 (Attn: BSC 23-29), Sears Tower, Chicago, Il 60684.

6.5 Changes from previous issue The margins of the specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in theses notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-21883D(NU)	CLOTH, SATIN OR CLOTH TWILL, RAYON WARP AND COTTON FILLING

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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